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Faculty of Applied Engineering and Computing

Bachelor of Information and Communication Technologies

Graduate Diploma in Information and Communication Technologies

Database Administration

BCDE214

Assignment One

Semester Two, 2026

Due date: 30 August 2026

Time: 5.00pm

Instructions:

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TOTAL MARKS: **100**

Student Name/ID

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Submissions received late will be subject to a penalty of 10% of the student's mark per working day.

This assignment is worth **25%** of the total marks for this course.

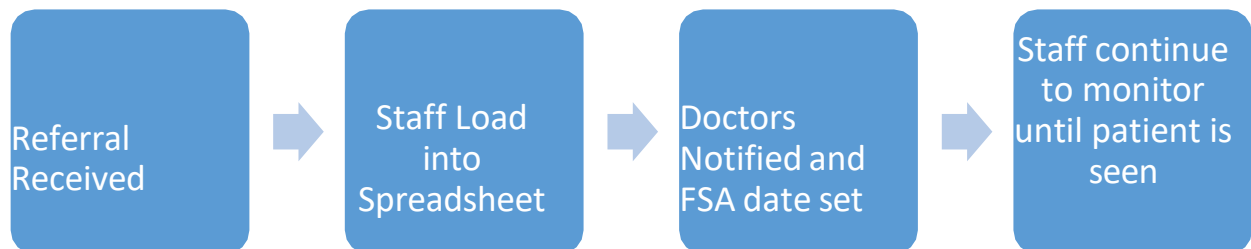
This paper has **seven** (7) pages including the cover sheet.

Aims and Instructions

- The aim of this assessment is for students to demonstrate their ability to evaluate, select and utilise techniques and tools used in the design and development of complex database projects.
- Apply knowledge to install, use and configure/reconfigure databases and perform reverse and forward engineering.
- Apply knowledge to investigate problems, write and test queries, and create reports.
- There will be time set aside during normal class time during which students will be asked to demonstrate some aspects of their assignment.
- Microsoft Visio or a similar tool must be used to create the Entity Relationship diagram. The Visio/pdf file and the MS SQL Server and MySQL database files along with the report need to be **submitted via Moodle** before the due date.

The Scenario

Sysmex Hospital receives thousands of referrals a year for complex surgeries. These are currently managed by a team of manual data trackers who sit in a darkened room and maintain spreadsheets using manual methods which are labour intensive. The process is roughly as follows:



Problem: Referrals are manually logged in an excel spreadsheet (**ARA July Data Wait Lists 2026.xlsx**) and are tracked on daily basis for changes using written memos and verbal orders. The management team of Sysmex Hospital have decided that all manual databases are to be moved into better solutions to allow automated population of the data and some reporting. In order to convince the Medical teams this will still be able to track patients adequately, a sample of referrals will be loaded in to a relational database and some reporting will be tested. For this purpose MySQL database has been selected and you have been commissioned to implement the project.

Task: Please place the data in the spreadsheet (**ARA July Data Wait Lists 2026.xlsx**) into a relational database. The fields from existing spreadsheet is listed below:

Datapoint	Simple Definition
Referral Date	Date the Referral was sent
Referred By	Person Sending Referral
NHI	Personal Identifier
Patient Name	Name specified by patient
DOB	Date of Birth
Department	Hospital Department
Added to Waitlist Date	Date referral was added to waitlist
Surgeon	Surgeon Name
FSA Date	First Specialist Appointment Date
Health Target Eligible	If referral is eligible for Ministry of Health Reporting

In addition the Medical Staff would ideally like to have these fields:

Patient Age at Referral
Days Waiting from Referral Date

Assumptions:

- We receive all referrals on the day they are sent
- Data has not been checked for quality but merely typed into the system
- The Ministry of Health expect everyone to be seen within 80 days
- Any reporting should exclude people who are not eligible for a health target

Desired Outcomes

A system that can be used to:

- Log contacts, including patient information, person who made contact information, worker information and actions taken. Note that actions taken may generate another contact.
- Store patient information, including who may have made contact on their behalf.
- Store worker information. This should also include when the worker was available for contact, as part of the reporting outcome may be requests for information as to why certain workers may not have had contacts directed to them.
- Export information to other systems in the health care sector, especially the palliative care sector. It is not known at this stage what format this data needs to be in.

Constraints

Constraints to the potential system include, but are not limited to the following:

- The patients are often rural and/or elderly. Digital literacy may not be high.
- Internet and cell phone coverage for patients cannot be guaranteed.
- Workers may require extensive training with any IT systems.
- Due to privacy concerns, real data cannot be provided. This is a mix of an old and new system.

Produce (please refer to the detailed marking rubrics):

1. A relational Schema (an extended ERD), using Visio, that includes entities, relationships, cardinality, attributes and the PK indicated.
2. A data dictionary of normalized tables (fields, data type, size, constraints)
3. A report on the GDPR compliant design process and the design issues encountered and how you chose to resolve these e.g. choice of entities, relationships, choice of keys, extent of normalization, multi-valued or composite attributes etc. Comment on any optional, mandatory, recursive, weak, subtype/supertype and composite entities.
4. Maintain and provide evidence of version control throughout the assignment.
5. Tables and relationships created in MySQL – with sample data entered (please refer to **ARA July Data Wait Lists 2026.xlsx** file) and *we would like to be able to answer the following queries.*
 - How many people have been referred for cardiothoracic?
 - What is the average time taken (in days) to see a Surgeon by Department?
 - Who has each Surgeon on their list and how long have they been waiting, or did they wait?
 - If all patients under 18 need to be seen by Pediatric Surgery, are there any patients who need to be reassigned?
 - What percentage of patients were seen within the target of 75 days by department?

BCDE 214 - Assignment 1 - Marking Rubrics

Name _____ Iterations: 1 2 3 Mark ____/100

1. General Data Protection Regulation (GDPR) [10 marks]

The General Data Protection Regulation (GDPR) is a new European Union (EU) data privacy law. It came into full effect on 25 May 2018.

The GDPR's main purpose is to create one coherent data protection framework across the EU, greatly improving data protection and privacy rights. It imposes a comprehensive set of principles and obligations which agencies working in or with the EU will need to be aware of and comply with.

1a Use the resources provided in Moodle to learn more about the GDPR and how it could affect your Health Tech Database solution. [5 Marks]

1b How well does your solution comply with data protection law? [5 Marks]

2. Design – Normalization and ERD [20 marks]

2a An Entity Relationship diagram of the system that you work on in an iteration. Use normalization and other design practices to an adequate standard.

Comments expected on:

Data Cleansing

Choice of Entities

Choice of attributes (simple, composite, multi-valued etc.) Choice of keys

Connectivity/Relationships

Use of composite/bridging entities Extent of normalization

Other “interesting” aspects of the design (e.g. supertype/subtype, cardinality – optional or mandatory, weak entity, recursive entity etc.)

Marks will be deducted for:

Attributes in wrong tables Incorrect PKs

Incorrect FKs Incorrect

cardinalities Incorrect entities

Relationships not shown

Not enough entities (two extra fields not created as requested) Improper

Normalization

3. Coding (Initial - Using MS SQL Server) [10 Marks]

3a Create initial database, Load data using code (no manual insert) and fix data errors

4. Coding (Final Implementation in MySQL and writing Queries) [35 Marks]

4a Final functioning Health Databases Application (Final version using MySQL Database) and associated files that meet the brief and user expectations with final backup and complete Data dictionary. Database match data dictionary/ERD and if required extra fields are to be derived. Features include Back up- Restore, Security and Role Management. A report along with sample scripts covering the above topics in relation to database implementation. A demo or show and tell required.

(25 marks)

4b Queries should be sensible, optimized and spitting desired output. A demo or show and tell required along with the screenshots and code.

(10 marks, 2 marks per query)

5. Timeline [5 Marks]

Estimated and Final showing task breakdown, order of tasks, estimated time, actual time, actual dates, variance between estimates and actuals. (Initial and final)

6. Database Incremental Development [4 marks]

Version Control – Initial and final

7. Functional Testing [6 marks]

Full range of testing of various deliverables. Show evidence, analyze the correctness and validity of tests and document them in your report.

8. Project Final Report [10 marks]

Produce a document integrating all document sections and appropriately formatted. Identify what worked / didn't work, and why, and what they would do differently next time.

Fully discusses the design, development and performance of the solution, including ethical and responsible use of AI in the assessment.

Expressed clearly and fluently, using specific examples.

Guide to preparing your report:

The topics in the Marking Rubrics on the previous page will be some of the content of your report. The report may also contain such items as screen snips of SQL code and the results from running SQL code. A detailed marking schedule/rubrics has also been shared with the assignment.

The report must address the topics above in the order that they are listed and use the topic headings as they are written.

The report will be one of the items to be submitted to the drop box **ALONG WITH** other files such as the ERD and SQL scripts etc.

Failure to follow these guidelines will potentially cause marks to be lost.